My long-term vision is to develop AI systems that remain reliable and effective when the data they rely on is complex, heterogeneous, and imperfect, the norm rather than the exception in most real-world settings. AI has advanced at a remarkable pace, from convolutional networks to Transformers and, most recently, to vision-language and large language foundation models, and these systems are now moving from benchmarks into real-world, high-stakes decisions. Yet key problems remain. **Structure:** models trained with pixel-wise objectives can be highly accurate yet structurally wrong (e.g., a broken connection or a spurious loop). **Trust:** models are often overconfident and do not know what they do not know. **Multimodality and data:** real-world data is heterogeneous, noisy, weakly labeled, or missing entire modalities. **Foundation models:** VLMs and LLMs must reason faithfully and efficiently over very large inputs, with calibrated confidence.

My research lies at the intersection of *machine learning* (**ML**), *computer vision* (**CV**), and *topological data analysis* (**TDA**), and follows one principle: **principled learning, which combines the flexibility of deep learning with principled priors from mathematics and statistics**. For *structure*, I use **algebraic topology** (persistent homology, discrete Morse theory, homotopy) so that predictions are topologically correct and models learn structural representations. For *reliability*, I build on **probability and statistics** to quantify uncertainty and calibrate confidence. *Interpretability* follows from both: when an output carries explicit structure and quantified uncertainty, users see not only what the model predicts, but how confident it is and where it may be wrong. My work is organized into four themes: **(1) Topological Deep Learning**, **(2) Trustworthy Machine Learning**, **(3) Multimodal AI and Generative AI**, and **(4) VLM/LLM Reasoning, Efficiency and Confidence**, applied across segmentation, registration, data synthesis, multimodal learning, and foundation-model reasoning, and published at NeurIPS, ICLR, CVPR, ICCV, and ECCV.

Research Progress

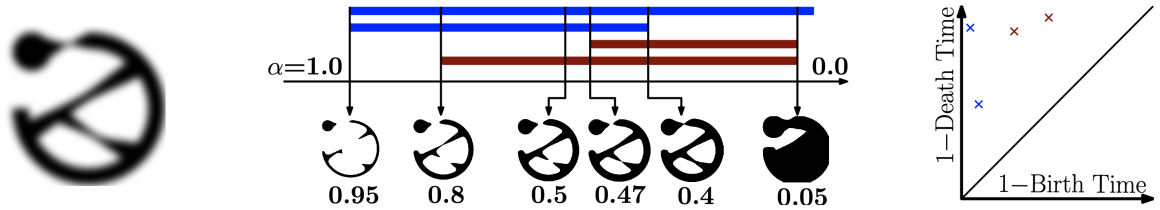
1 Topological Deep Learning. Deep networks mostly learn pixel-wise representations with pixel-wise losses, which are blind to global structure: a few misclassified pixels can break a thin connection or merge two objects, barely changing accuracy but completely changing the structure of the output. *How can we learn topological representations to understand the topology, structure, and geometry of images?*

> **As a pioneer in topological deep learning**, I have designed topology-preserving deep learning algorithms [1, 2, 3, 4, 5, 6, 7, 8, 9, 10], and leveraged topological tools to reason about structures directly [11, 12].

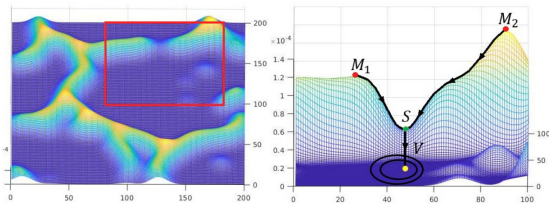
Differentiability of topological tools. Topological quantities, such as Betti numbers (the number of connected components and loops), are discrete and not directly differentiable, the key obstacle to using them in end-to-end training. In my NeurIPS'19 work [1], among the first to bring persistent homology into deep network training, I proposed a *topological loss* that measures the distance between the persistence diagrams of the predicted likelihood map and the ground truth (Fig. 1a). The key insight is that each point in a persistence diagram is determined by a pair of *critical pixels* (extrema and saddles) of the likelihood; back-propagating through them yields a well-defined gradient that fixes exactly the locations responsible for topological errors. To make this efficient and scalable to 3D, I proposed a loss based on *discrete Morse theory* [2] (ICLR'21, Spotlight) that identifies entire critical structures rather than individual pixels (Fig. 1b), and *homotopy warping* [3] (NeurIPS'22), which uses digital topology to locate critical pixels by warping the prediction toward the ground truth (Fig. 1c). With collaborators, I extended topological priors to 3D [6, 7], to multi-class topological interactions enforced efficiently by convolution [8] (ECCV'22, Oral), to semi-supervised learning [5, 10], and beyond segmentation, to Trojan detection, where a topological prior recovers compact, connected trigger patterns [9] (ICLR'22).

Deep structural reasoning. Beyond topology-preserving prediction, I proposed *a deep learning method that directly learns the topological/structural representation of images* [11] (ICLR'23, Spotlight). Using discrete Morse theory, an image is decomposed into a Morse complex of structural elements, such as branches and patches, and the network learns a probabilistic model over them. This moves inference from pixel space to structure space and enables *structure-level* uncertainty: an annotator can inspect and accept or reject an entire branch instead of individual pixels. The ideas apply broadly to thin, curvilinear, and networked structures, such as vessels, neurons, membranes, and road networks.

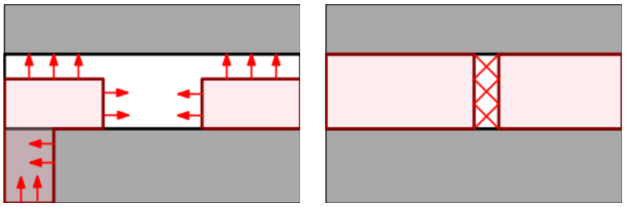
2 Trustworthy Machine Learning: Reliability, Interpretability, and Robustness. Deep networks



(a) Persistent homology: filtration, barcode, and persistence diagram [1]

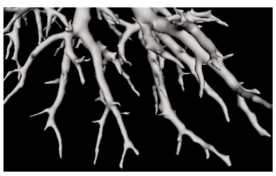


(b) Discrete Morse theory [2, 11]

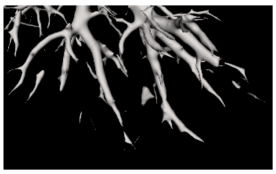


(c) Homotopy warping [3]

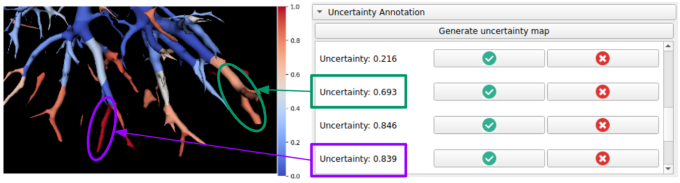
Figure 1: Classical topological tools made differentiable for deep learning: (a) birth/death of topological features is determined by critical pixels; (b) critical points define Morse structures; (c) topology-preserving warping exposes critical pixels.



(a) Ground truth



(b) Prediction



(c) Structure uncertainty (d) Interactive 3D Slicer UI

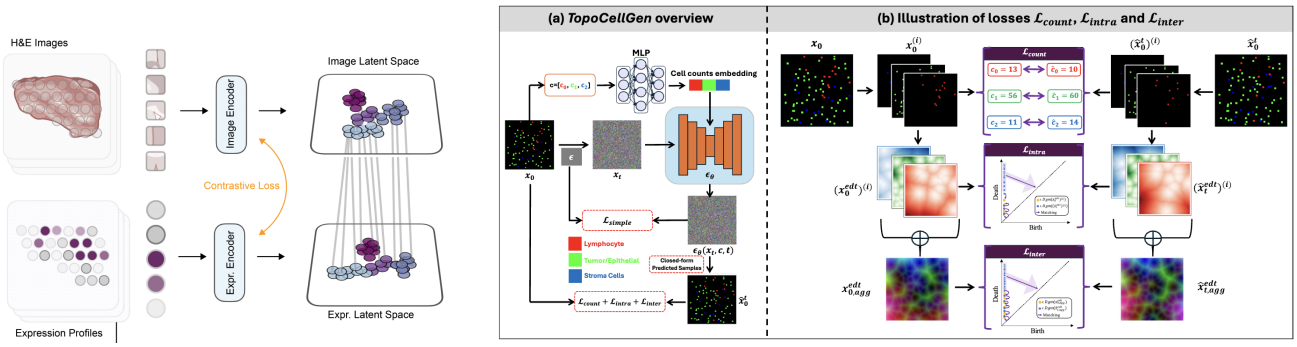
Figure 2: Structure-level uncertainty for human-in-the-loop annotation [12, 13]: per-branch uncertainty (c) is exposed in an interactive 3D Slicer interface (d), so annotators verify only uncertain structures.

are often overconfident; in high-stakes settings, identifying low-confidence samples or regions tells us where a model should defer to a human. Meanwhile, ideal training data is rarely available. *How can we build models whose confidence is reliable and whose outputs are interpretable, even when the data is imperfect?*

I have developed principled uncertainty estimation for structured outputs [11, 12, 13], unlabeled data [14, 15], and image registration [16, 17], and robust learning under limited or noisy labels and missing modalities [15, 18, 5, 19].

Reliability and interpretability. *Structure-level uncertainty.* Pixel-wise uncertainty highlights nearly every boundary pixel and offers limited guidance. Building on the Morse-complex representation, my work estimates uncertainty per structure [11], and follow-up work models dependencies among structures with a graph neural network [12] (NeurIPS'23). We integrated structure-level uncertainty into the open-source 3D Slicer platform for interactive annotation, where annotators click on a branch to see its uncertainty and decide whether to keep it (Fig. 2) [13]. *Confidence without labels.* I co-developed a method that uses the consistency of predictions during training on unlabeled data to learn a confidence estimator with a ranking objective [14] (ICLR'23), improving active learning, and calibrated uncertainty to select reliable pseudo-labels for semi-supervised counting [15] (ICCV'23). *Uncertainty for dense correspondence.* Registration errors silently propagate to every quantity computed from the deformation. I proposed a hierarchical framework that propagates uncertainty from individual correspondences to the transformation and to downstream predictions [16] (ICLR'25), building on interpretable registration by regression [20]. Because many models are already deployed, we further proposed an inference-time, model-agnostic method that estimates uncertainty for *any pretrained* registration network without retraining, by exploiting transformation equivariance [17] (NeurIPS'26).

Robustness. I have developed algorithms that learn from weak or limited supervision, including anomaly-guided weakly supervised segmentation [18], calibrated pseudo-labeling [15], and noise-aware topological consistency [5, 10]; meta-learning of modality-agnostic representations that remain accurate when input channels are missing at test time [19] (ICCV'23); and segmentation that scales to ultra-high-resolution 3D volumes [21].



(a) Cross-modal alignment [23] (b) Topology-aware diffusion [25]

Figure 3: Multimodal and generative AI. (a) Aligning image and non-visual latent spaces with a contrastive, ranking-consistent objective. (b) A diffusion model trained with count and persistence-diagram-based losses to generate layouts with correct spatial topology.

3 Multimodal AI and Generative AI (GenAI). Complementary signals (different imaging contrasts, paired visual and molecular measurements, heterogeneous sensors) are often available but underused, and generative models are increasingly used to synthesize, complete, or translate such data. *How can we learn representations robust across modalities, and generative models that respect the structure of the data they generate?*

I have developed multimodal representation learning robust to missing or varying modalities [19, 22, 23], and generative models that produce topologically or physically consistent outputs [24, 25, 26].

Multimodal representation learning. Beyond meta-learning for missing modalities [19], Brain-ID [22] learns, from synthetic data, contrast-agnostic features that are stable across large variations in acquisition settings. RankByGene [23] (TMI'26) aligns image features with paired non-visual (gene-expression) signals through a cross-modal ranking-consistency objective (Fig. 3a), avoiding the need for exact pointwise correspondence between modalities.

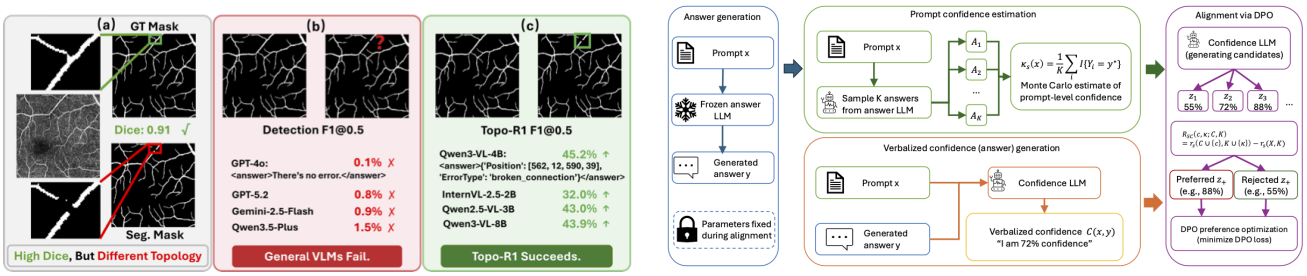
Structure-aware generative modeling. Standard GANs and diffusion models match a data distribution without constraints on structural correctness, a critical failure mode when generated content is later counted or measured. I proposed TopoCellGen [25] (CVPR'25, Oral), a diffusion model that enforces correct spatial and topological arrangements of generated object layouts via persistence-based losses (Fig. 3b), and a diffusion-based domain adaptation method with multi-channel, uncertainty-weighted score matching [26] (ECCV'26). Learn2Synth [24] (ICCV'25) learns the data-synthesis strategy itself: instead of a hand-designed augmentation pipeline, it differentiates through a downstream model's performance on real data (hypergradients) to optimize what the synthesizer should produce. We also showed that unrolled networks for inverse problems can be interpreted as conditional probability flows [27].

4 VLM/LLM Reasoning, Efficiency and Confidence. VLMs and LLMs inherit, at a much larger scale, the questions that drive my work: *Can a model reason over the parts of an input that matter, and does it know what it knows?*

I have developed selective and structure-aware reasoning for VLMs [28, 29], efficient multimodal LLMs via learned token compression [30], and calibrated verbalized confidence for LLMs [31].

Reasoning and efficiency. Gigapixel images far exceed any VLM's context: uniform downsampling discards the fine detail that matters, while exhaustive tiling is prohibitively expensive. HistoSelect [28] (CVPR'26) mimics how a human expert inspects such an image, first selecting a small set of informative regions and then reasoning only over them; LoC-Path [30] learns to compress redundant visual tokens fed to a multimodal LLM. Topo-R1 [29] brings topology to foundation models: general-purpose VLMs fail to perceive topological errors (e.g., broken connections) even when pixel overlap is high, and we equip them to detect and localize such anomalies (Fig. 4a).

Confidence. LLMs increasingly state their confidence in natural language, but this verbalized confidence is often misaligned with correctness. ORCE [31] first estimates prompt-level confidence by sampling, then aligns the model's verbalized confidence via preference optimization so that its ordering



(a) Topo-R1: topological anomaly detection [29] (b) ORCE: aligning verbalized confidence [31]

Figure 4: Trustworthy foundation models: structure-aware visual reasoning in VLMs (a) and calibrated, verbalized confidence in LLMs (b).

reflects actual correctness (Fig. 4b), echoing the ranking view of confidence in [14].

Ongoing and Future Work

Building on this foundation, I plan to pursue four directions:

- ***Learning with real-world (imperfect) data.*** Most real-world data is weakly labeled, partially labeled, noisy, or unlabeled. I will develop methods that exploit the rich information in unlabeled data (weakly, semi-, and self-supervised learning) and close the loop between a model's uncertainty and the supervision that would most improve it, e.g., active learning and human-in-the-loop proofreading driven by structure-level uncertainty (Fig. 5), and study how structural accuracy and uncertainty propagate to downstream analysis. More broadly, I will ask whether a model can learn *what data it should be trained on*, jointly with the model itself, connecting to data selection and valuation for large models.

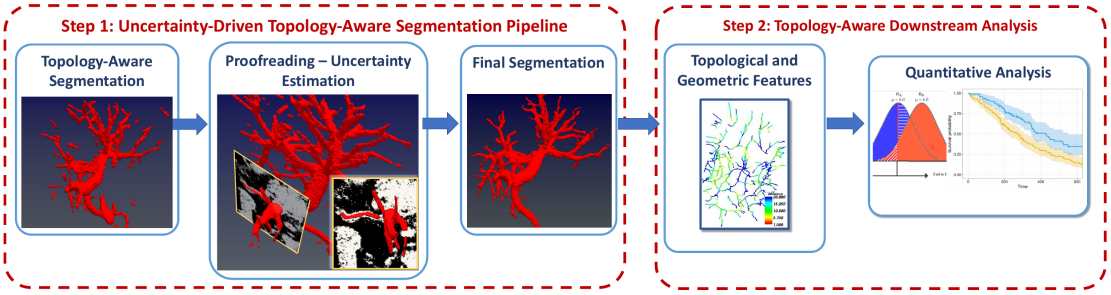


Figure 5: Ongoing work on learning with imperfect data: uncertainty-driven, topology-aware prediction with human proofreading of uncertain structures (Step 1), followed by topology/geometry-aware downstream analysis (Step 2).

- ***Multimodal AI.*** The growing availability of paired images, text, tabular records, and other measurements calls for methods that align and leverage different modalities for downstream tasks. I will develop alignment objectives that do not require exact correspondence, representations that remain robust when modalities are missing at test time, and principled ways to quantify how much each modality contributes.
- ***Generative AI.*** Synthetic data can address data scarcity, but only if it is faithful. I will integrate topological and geometric priors directly into diffusion and flow-based models, so that structural correctness is a property of the generative process rather than a post-hoc filter, and learn synthesis strategies end-to-end from downstream objectives, as in Learn2Synth.
- ***Trustworthy foundation models.*** I will make VLMs and LLMs reason *faithfully*, *efficiently*, and with *calibrated confidence*: structure-aware visual reasoning (perceiving connectivity, loops, and spatial relations), efficient reasoning over gigapixel-scale inputs, and calibrated, verbalized confidence, unified in a single framework rather than bolted on separately. This direction brings together my two main lines of work, topology and trustworthy ML.

I will continue to develop AI systems that are structure-aware, reliable, and efficient, building on methods already validated across a wide range of real-world imaging problems [18, 32, 33, 34, 35, 21], and I look forward to collaborating across machine learning, computer vision, computational topology, and application domains in science and engineering.

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